

SIMATS ENGINEERING

ASSESSMENT TOOL 1

COURSE NAME: CSA16 **COURSE CODE:** Data Warehousing and Data Mining

COURSE OUTCOME COVERED

CO4: Examine the applicability of clustering algorithms in real-world scenarios, presenting findings through clear reports with effective visualizations.

QUESTIONS: 05

Total Marks:50

Annexure A

Data Interpretation

Criteria	Understanding of Data (2.5)	Analysis (2.5)	Application of Concepts (2)	Conclusion (1.5)	Clarity (1)	Total(10)
Weightage	25%	25%	20%	20%	10%	100%
<i>Q1</i>						
<i>Q2</i>						
<i>Q3</i>						
<i>Q4</i>						
<i>Q5</i>						

Code of Conduct:

I _____ (Name / Reg No) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment.

I understand that any violation of academic integrity rules will result in disciplinary action.

Signature of the student:

RUBRICS FOR CALCULATION:

Criteria	Weightage	Level 4 (Exemplary)	Level 3 (Proficient)	Level 2 (Developing)	Level 1 (Beginning)
Understanding of Data	25%	Accurately interprets all data patterns, trends, and relationships	Interprets data correctly with minor gaps	Partial understanding; misses key trends	Misinterprets or fails to understand data
Analysis	25%	Provides deep analysis with strong logical reasoning and justification	Provides reasonable analysis with some justification	Basic analysis with weak reasoning	No meaningful analysis provided
Application of Concepts	20%	Effectively applies relevant concepts/tools to interpret data accurately	Applies concepts with minor errors	Limited application; requires guidance	Unable to apply concepts to data
Conclusion	20%	Draws clear, meaningful conclusions with strong insights and implications	Provides valid conclusions with moderate insights	Conclusions are vague or weak	No clear or valid conclusions
Clarity	10%	Presents interpretation clearly using proper structure, visuals, and terminology	Generally clear with minor issues	Lacks clarity or proper structure	Poor presentation and difficult to understand

Annexure A

Q1. Dataset: Hospital Patient Segmentation

Patient ID	Age	Monthly Medical Expense (₹)	Health Risk Score
P1	28	12000	65
P2	55	25000	40
P3	42	20000	75
P4	30	11000	70
P5	60	28000	35

SSE Values

K SSE

1	600
2	350
3	220
4	200
5	190

Question (BL4 – 8 Marks):

A healthcare organization uses K-Means clustering to group patients based on their medical expenses and health risk scores. Interpret the elbow point from the SSE values provided and determine the optimal number of clusters. Discuss how selecting this number of clusters affects patient segmentation and healthcare decision-making.

Q2. Dataset: E-Commerce Order Analysis

Order ID	Product Category	Order Value (₹)
O1	Electronics	55000
O2	Clothing	25000
O3	Electronics	57000
O4	Furniture	30000
O5	Clothing	27000

Question (BL4 – 8 Marks):

An online retailer applies hierarchical clustering to group customer orders based on product category and order value. A dendrogram is generated from the dataset. Interpret how the clusters are formed and identify the appropriate number of clusters. Explain the significance of the selected clusters in understanding customer purchasing behavior.

Q3. Dataset: Mobile App User Behavior

	User	Daily Usage Time (min)	Login Frequency	In-App Purchase (₹)
User Data	U1	120	18	2000
	U2	60	5	5000
	U3	140	22	1500
	U4	50	4	6000
	U5	130	20	1800

Question (BL5 – 8 Marks):
A mobile application company uses the HAC algorithm to segment users based on their activity patterns. Interpret how users are assigned to clusters using probability-based membership values. Discuss the advantages of soft clustering over hard clustering for customer behavior analysis.

Q4. Dataset: Agricultural Crop Monitoring

	Farm	Rainfall (mm)	Soil Fertility Score	Crop Yield (tons)
F1	300	8	85	
F2	500	4	70	
F3	400	6	80	
F4	280	9	90	
F5	520	3	65	

Question (BL4 – 8 Marks):
An agricultural research center performs clustering on farm data before and after applying normalization. Interpret the differences observed in the clustering results. Explain how normalization influences cluster formation and improves the reliability of agricultural data analysis.

Q5. Dataset: Employee Performance Evaluation

Employee Productivity Score Attendance (%) Skill Assessment Score			
E1	88	92	84
E2	62	75	68
E3	78	88	72
E4	55	65	58
E5	93	96	90

Silhouette Scores

Model Score

K=2 0.48

K=3 0.68

K=4 0.52

Question (BL5 – 8 Marks):

A company clusters employees based on productivity, attendance, and skill assessment scores. Analyze the silhouette scores obtained for different clustering models and determine the most suitable number of clusters. Justify your answer and explain how the selected model contributes to effective employee performance evaluation.

CSA16 – DATA WAREHOUSING AND DATA MINING

CO4 – ANSWERS

Q1. Hospital Patient Segmentation – K-Means and Elbow Method

Given SSE values: K=1: 600, K=2: 350, K=3: 220, K=4: 200, K=5: 190.

Interpretation: The SSE decreases sharply from 600 to 350 when K changes from 1 to 2, and again to 220 for K=3. After K=3, the improvement becomes very small: only 20 for K=4 and 10 for K=5. Therefore, the elbow point is at **K=3**.

Optimal number of clusters: 3 clusters.

Effect on patient segmentation: Three clusters can meaningfully separate patients into groups such as: (1) lower medical expense with higher health risk, (2) medium expense with moderate risk, and (3) higher expense with lower health risk. For the given data, patients P1 and P4 show relatively low expenses and high risk, P3 is intermediate, and P2 and P5 have higher expenses and lower risk.

Healthcare decision-making: This segmentation helps the organization prioritize high-risk patients, plan targeted treatment programs, allocate medical resources, estimate costs, and design suitable preventive-care strategies. Selecting more than three clusters would add complexity with only a small reduction in SSE, while fewer clusters may combine patients with different health needs.

Conclusion: Based on the elbow method, **K=3** provides a good balance between compact clusters and model simplicity.

Q2. E-Commerce Order Analysis – Hierarchical Clustering

Dataset pattern: O1 and O3 are Electronics orders with very high values of ■55,000 and ■57,000. O2 and O5 are Clothing orders with lower values of ■25,000 and ■27,000. O4 is a Furniture order with a value of ■30,000.

Cluster formation: In hierarchical clustering, the most similar orders are merged first. O1 and O3 are highly similar because they belong to the same category and have close order values. O2 and O5 also form a natural cluster for the same reason. O4 can remain separate initially because its product category differs, or it can join the lower-value group at a higher linkage level depending on the encoding and distance measure used.

Appropriate number of clusters: A meaningful interpretation is **3 clusters**: High-value Electronics, Lower-value Clothing, and Furniture. The dendrogram cut should be selected at a level that preserves these major purchasing patterns.

Significance: The clusters reveal different customer purchasing behavior. High-value Electronics orders indicate premium or technology-oriented spending, Clothing orders represent lower-value routine purchases, and Furniture represents a separate product preference. The retailer can use these groups for personalized recommendations, targeted promotions, inventory planning, and customer-value analysis.

Conclusion: Hierarchical clustering provides an interpretable view of order similarity, and **3 clusters** best highlights the main product and spending patterns in this small dataset.

Q3. Mobile App User Behavior – Soft Clustering

Interpretation of user groups: U1, U3, and U5 show high daily usage and frequent logins, with relatively lower in-app purchase values. U2 and U4 show lower usage and login frequency but higher purchase values.

Probability-based membership: In soft clustering, a user is assigned membership probabilities for multiple clusters rather than being placed in only one cluster. For example, a highly active user may have a high probability of belonging to an “Active User” cluster and a smaller probability of belonging to a “Moderate Spender” cluster. Similarly, a user with high purchases but low usage may strongly belong to a “High-Value Purchaser” cluster while still having some membership in other groups.

Advantages over hard clustering:

1. It represents uncertainty in real customer behavior.
2. Users can show characteristics of more than one segment.
3. It supports more personalized recommendations and marketing.
4. It avoids forcing borderline users into a single group.
5. Membership values help identify users whose behavior is changing over time.

Conclusion: Soft clustering is more suitable for customer behavior analysis because mobile-app users often have overlapping activity and spending patterns. It provides richer information than a strict one-cluster assignment.

Q4. Agricultural Crop Monitoring – Effect of Normalization

Before normalization: The features are measured on different scales. Rainfall values are around 280–520, soil fertility scores are 3–9, and crop yield values are 65–90. A distance-based clustering algorithm may therefore be strongly influenced by rainfall because it has the largest numerical range.

After normalization: Each feature is scaled to a comparable range or distribution. This prevents rainfall from dominating the distance calculation and gives soil fertility and crop yield appropriate influence in cluster formation.

Observed clustering pattern: F1 and F4 are similar because they have lower rainfall, higher soil fertility, and higher crop yield. F2 and F5 are also similar because they have higher rainfall, lower fertility, and lower yield. F3 represents an intermediate farm profile. After normalization, these relationships can be identified more reliably because all variables contribute more fairly.

Reliability improvement: Normalization improves the fairness of distance calculations, reduces scale bias, produces more meaningful clusters, and supports better agricultural decisions such as identifying high-performing farms, low-fertility areas, and farms requiring intervention.

Conclusion: Normalization is essential when variables have different units and ranges. It improves the quality and reliability of agricultural clustering results.

Q5. Employee Performance Evaluation – Silhouette Analysis

Given silhouette scores: K=2: 0.48, K=3: 0.68, K=4: 0.52.

Analysis: The silhouette score measures how well data points fit within their own cluster and how clearly they are separated from other clusters. A higher score indicates better-defined and better-separated clusters. The highest score is **0.68 for K=3**.

Most suitable number of clusters: 3 clusters. K=2 provides only moderate separation, while K=4 reduces the score and may divide the employee data into unnecessary or less distinct groups.

Performance interpretation: The three clusters can represent groups such as high performers, average performers, and employees needing improvement or support. For example, E1 and E5 show strong

productivity, attendance, and skill scores; E2 and E4 show lower values; E3 lies between these groups.

Contribution to effective evaluation: The selected model helps management identify high performers for recognition, average performers for skill development, and lower-performing employees for focused training and support. It also supports data-driven performance evaluation rather than relying only on subjective judgment.

Conclusion: Since **K=3 has the highest silhouette score of 0.68**, it is the most suitable clustering model for this dataset and provides the clearest employee performance segmentation.